

Results from fitting to reference p-curves

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::: {.callout-note title = "Background"} While building the simulations (before knowing the actual target p-curve by Clarke et al.), to test our fitting procedure, we used three existing empirical p-curves as targets:

1. **Sotola** ($k = 1,556$ significant p-values): Sotola (2023), Meta-Psychology. [doi](#) → [results](#)
2. **Wetzels** ($k = 593$): Wetzels et al. (2011), Perspectives on Psychological Science. [doi](#) → [results](#)
3. **Simonsohn** ($k = 22$): Simonsohn et al. (2014), JEP: General, Figure 3B. [data](#) → [results](#)

In this document, we report the results of the best-fitting simulations for each of those empirical p-curves including: i) visualisations of the top 5 best fitting simulations to each empirical p-curves for each world; ii) fit indices for the best simulations in each world for each empirical p-curve; iii) the parameter values for the top 10 best fitting p-curves (across all worlds) for each empirical p-curve. :::

Set-up code: Data wrangling and goodness of fit calculations

```
# Combine simulation results from all conditions into one big dataset
devtools::load_all("../..") # load fit p curve with updated plot_pcurves() function
library(rio)
library(tidyverse)
library(ggplot2)
library(kableExtra)

# reference p-curves
pcurves <- import("reference-pcurves.csv")

# load the multiple DV simulation results:
```

```

simres_realistic <- import("../sim-results/sim_realistic.csv")
simres_realistic$condition <- "multDV_realistic"
simres_worst <- import("../sim-results/sim_worst.csv")
simres_worst$condition <- "multDV_worst"
simres_perfect <- import("../sim-results/sim_perfect.csv")
simres_perfect$condition <- "multDV_perfect"
simres_H0 <- import("../sim-results/sim_H0.csv")
simres_H0$condition <- "multDV_H0"

# Delete unnecessary rows for different reporting strategies from simres_perfect and sim
simres_perfect <- simres_perfect |> filter(strategy == 3)
simres_H0 <- simres_H0 |> filter(strategy == 3)

# load the optional stopping simulation results:
simres_optStop_worst <- import("../sim-results/sim_optStop_worst.csv")
simres_optStop_worst$condition <- "optStop_worst"
simres_optStop_realistic <- import("../sim-results/sim_optStop_realistic.csv")
simres_optStop_realistic$condition <- "optStop_realistic"

simres_multDV <- rbind(simres_realistic, simres_worst, simres_perfect, simres_H0)
simres_multDV$binsum <- rowSums(simres_multDV[, paste0("p", 1:5)])
stopifnot(all(simres_multDV$binsum |> round(6) == 1))
simres_multDV$type <- "multDV"

simres_optStop <- rbind(simres_optStop_realistic, simres_optStop_worst)
simres_optStop$binsum <- rowSums(simres_optStop[, paste0("p", 1:5)])
stopifnot(all(simres_optStop$binsum |> round(6) == 1))
simres_optStop$type <- "optStop"

# Combine simulations from both p-hacking strategies (they have different parameter columns)
simres <- bind_rows(simres_multDV, simres_optStop)

# Compute G-squared fit statistics
simres$fit_sotola_g2 <- g2(
  observed = simplify2array(pcurves[1, paste0("p", 1:5)]),
  expected = simres[, paste0("p", 1:5)], n = pcurves$nsig[1], tol=1e-6)

simres$fit_wetzels_g2 <- g2(
  observed = simplify2array(pcurves[2, paste0("p", 1:5)]),
  expected = simres[, paste0("p", 1:5)], n = pcurves$nsig[2], tol=1e-6)

simres$fit_simonsohn_g2 <- g2(

```

```

    observed = simplify2array(pcurves[3, paste0("p", 1:5)]),
    expected = simres[, paste0("p", 1:5)], n = pcurves$nsig[3], tol=1e-6)

# Also compute chi-squared and RMSE for robustness checks
# Chi-squared
simres$fit_sotola_chi2 <- chi2(
  observed = simplify2array(pcurves[1, paste0("p", 1:5)]),
  expected = simres[, paste0("p", 1:5)], n = pcurves$nsig[1], tol=1e-6)

simres$fit_wetzels_chi2 <- chi2(
  observed = simplify2array(pcurves[2, paste0("p", 1:5)]),
  expected = simres[, paste0("p", 1:5)], n = pcurves$nsig[2], tol=1e-6)

simres$fit_simonsohn_chi2 <- chi2(
  observed = simplify2array(pcurves[3, paste0("p", 1:5)]),
  expected = simres[, paste0("p", 1:5)], n = pcurves$nsig[3], tol=1e-6)

# RMSE
simres$fit_sotola_rmse <- rmse(
  observed = simplify2array(pcurves[1, paste0("p", 1:5)]),
  expected = simres[, paste0("p", 1:5)], n = pcurves$nsig[1])

simres$fit_wetzels_rmse <- rmse(
  observed = simplify2array(pcurves[2, paste0("p", 1:5)]),
  expected = simres[, paste0("p", 1:5)], n = pcurves$nsig[2])

simres$fit_simonsohn_rmse <- rmse(
  observed = simplify2array(pcurves[3, paste0("p", 1:5)]),
  expected = simres[, paste0("p", 1:5)], n = pcurves$nsig[3])

# Compute BICs for between-world comparisons (accounting for model complexity)

# Number of free parameters in each simulated "world"
simres$free_parameters <- recode_values(simres$condition,
                                         "multDV_realistic" ~ 7,
                                         "multDV_worst" ~ 4,
                                         "multDV_perfect" ~ 3,
                                         "multDV_H0" ~ 0,
                                         "optStop_realistic" ~ 7,
                                         "optStop_worst" ~ 4)

# Compute BIC from G-squared and number of free parameters

```

```

simres$BIC_sotola <- simres$fit_sotola_g2 + simres$free_parameters * log(pcurves$nsig[1])

simres$BIC_wetzels <- simres$fit_wetzels_g2 + simres$free_parameters * log(pcurves$nsig[1])

simres$BIC_simonsohn <- simres$fit_simonsohn_g2 + simres$free_parameters * log(pcurves$nsig[1])

# Save condition labels for later use
condition_labels <- c(
  multDV_perfect      = "Perfect World",
  multDV_H0           = "Flat World",
  multDV_realistic    = "Realistic World: Multiple DVs",
  multDV_worst        = "Worst World: Multiple DVs",
  optStop_realistic   = "Realistic World: Optional stopping",
  optStop_worst       = "Worst World: Optional stopping"
)

```

1. Target p-curve: Sotola (2023):

Visualisations of best simulations in each world

You can visually inspect the fit of the top 5 (where available) simulations in each world (in blue) to the target p-curve (in black). Note that the Flat World only involved one simulation.

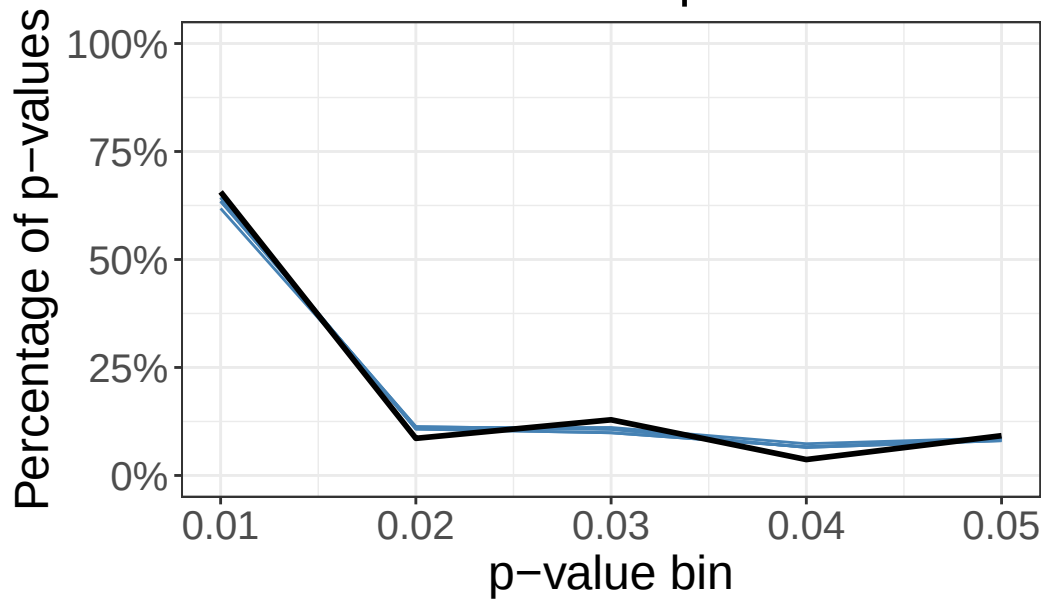
Realistic: Multiples DVs

```

plot_pcurves(simres |> filter(condition == "multDV_realistic"), poriginal = simplify2array(p

```

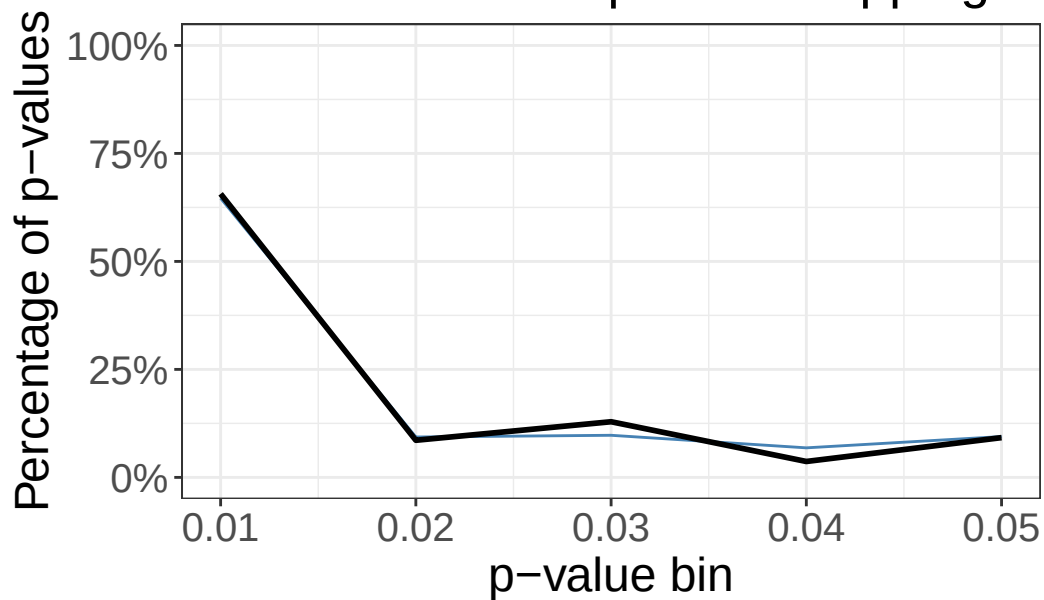
Realistic World: Multiple DVs



Realistic: Optional Stopping

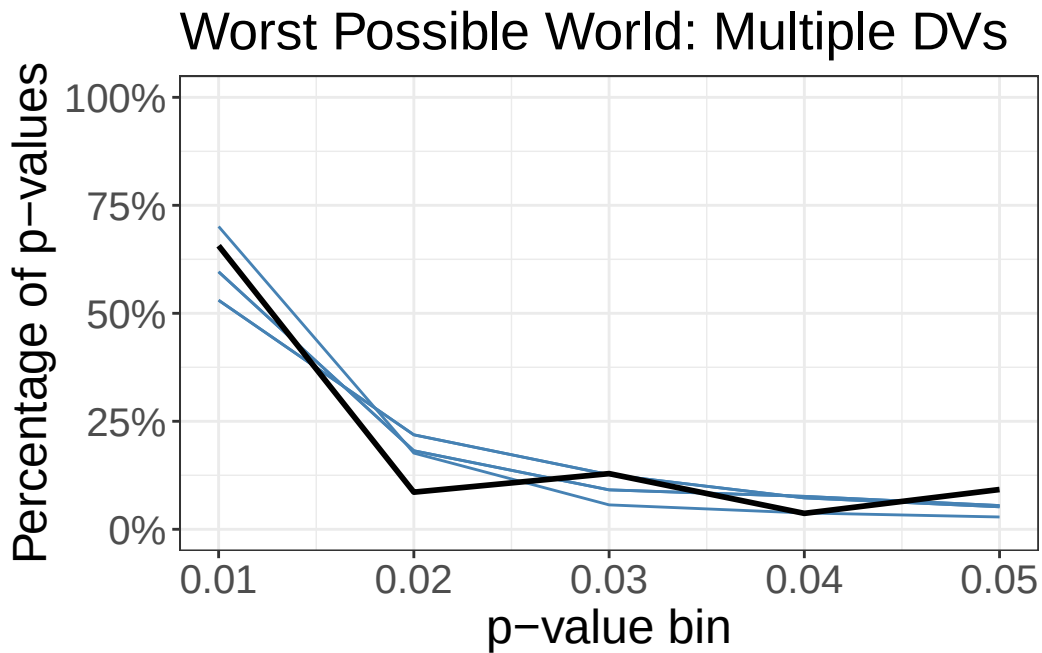
```
plot_pcurves(simres |> filter(condition == "optStop_realistic"), poriginal = simplify2array()
```

Realistic World: Optional Stopping



Worst: Multiple DVs

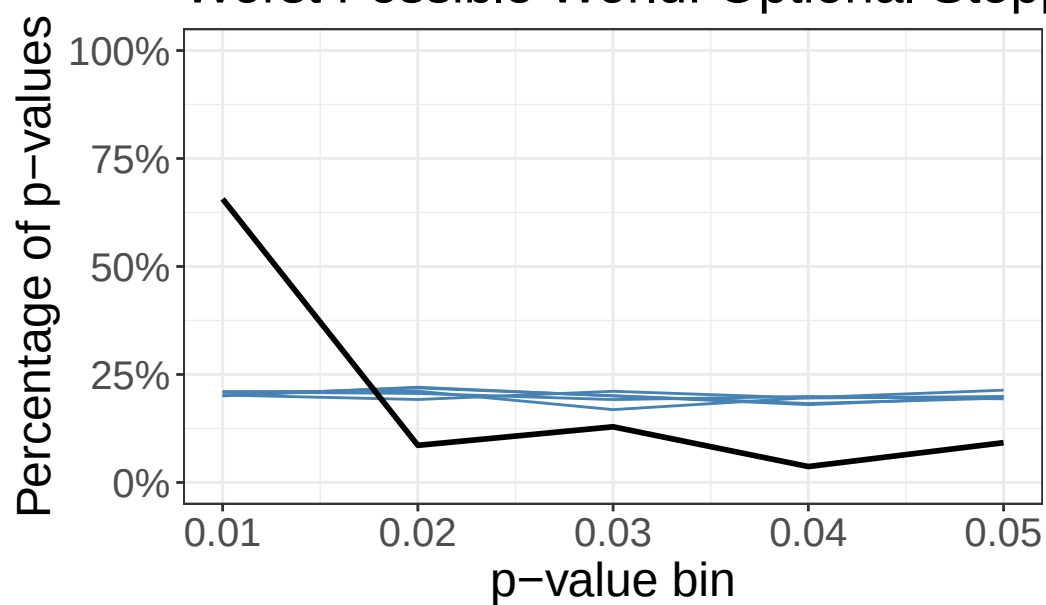
```
plot_pcurves(simres |> filter(condition == "multDV_worst"), poriginal = simplify2array(pcurv
```



Worst: Optional Stopping

```
plot_pcurves(simres |> filter(condition == "optStop_worst"), poriginal = simplify2array(pcurv
```

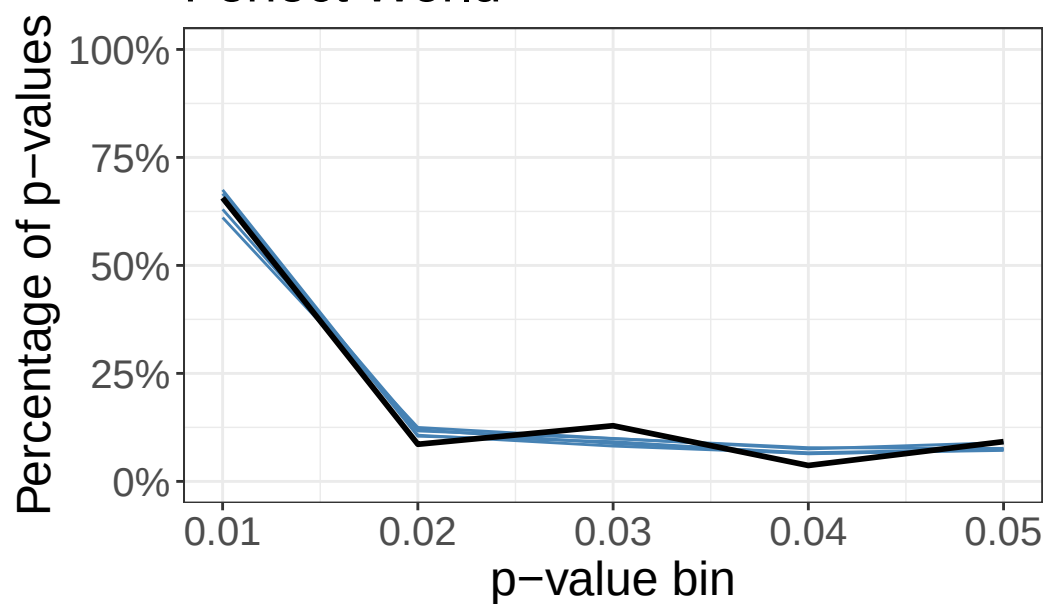
Worst Possible World: Optional Stopping



Perfect

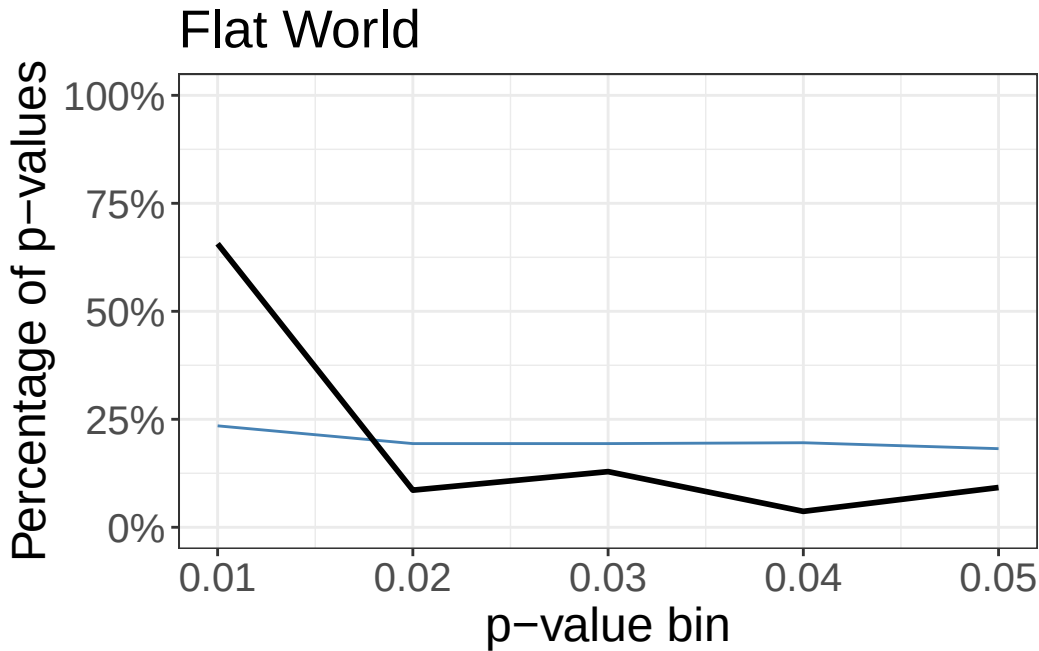
```
plot_pcurves(simres |> filter(condition == "multDV_perfect"), poriginal = simplify2array(pcu
```

Perfect World



Flat

```
plot_pcurves(simres |> filter(condition == "multDV_H0"), poriginal = simplify2array(pcurves[
```



Overview of best fitting simulations in each world

These fit indices reflect the fit of the best-fitting simulation in each condition/ world. Lower numbers indicate better fit. Note that BIC is a transformation of G2 that penalises for model complexity.

```
fit_tab <- simres %>%
  group_by(condition) %>%
  summarise(
    G2    = min(fit_sotola_g2),
    BIC    = min(BIC_sotola),
    Chi2   = min(fit_sotola_chi2),
    RMSE   = min(fit_sotola_rmse)
  )

fit_tab %>%
  mutate(condition = recode(condition, !!!condition_labels)) %>%
  mutate(across(
```

condition	G2	BIC	Chi2	RMSE
Flat World	139.56	139.56	164.99	0.21
Perfect World	5.62	20.90	5.57	0.02
Realistic World: Multiple DVs	4.55	40.21	4.14	0.02
Worst World: Multiple DVs	20.97	41.34	19.18	0.06
Realistic World: Optional stopping	4.50	40.16	4.17	0.02
Worst World: Optional stopping	160.54	180.91	200.30	0.23

```
-condition,
~ cell_spec(sprintf("%.2f", .x), bold = .x == min(.x))
)) %>%
kbl(escape = FALSE) %>%
kable_styling()
```

According to G2, the Realistic World: Optional stopping contained the best-fitting simulation to the target p-curve. Once model complexity is accounted for, the Perfect World contained the simulation with the best fit to the target p-curve. Results for chi-squared and RMSE are also provided as robustness checks.

What were the top 10 best fitting simulations (according to G^2)?

These rows reflect the 10 simulations that had the best fit to the target p-curve according to G^2 . (Scroll to see the other fit indices for these simulations.)

```
simres |> arrange(fit_sotola_g2) |> slice(1:10) |>
  select(condition, nvar, r, d, prop_Hacker, prop_H1, het, strategy, nmin, nmax, stepsize, c
  kbl(digits = 2) |>
  kable_styling() |>
  scroll_box(width = "100%", height = "400px")
```

2. Target p-curve: Wetzels (2011):

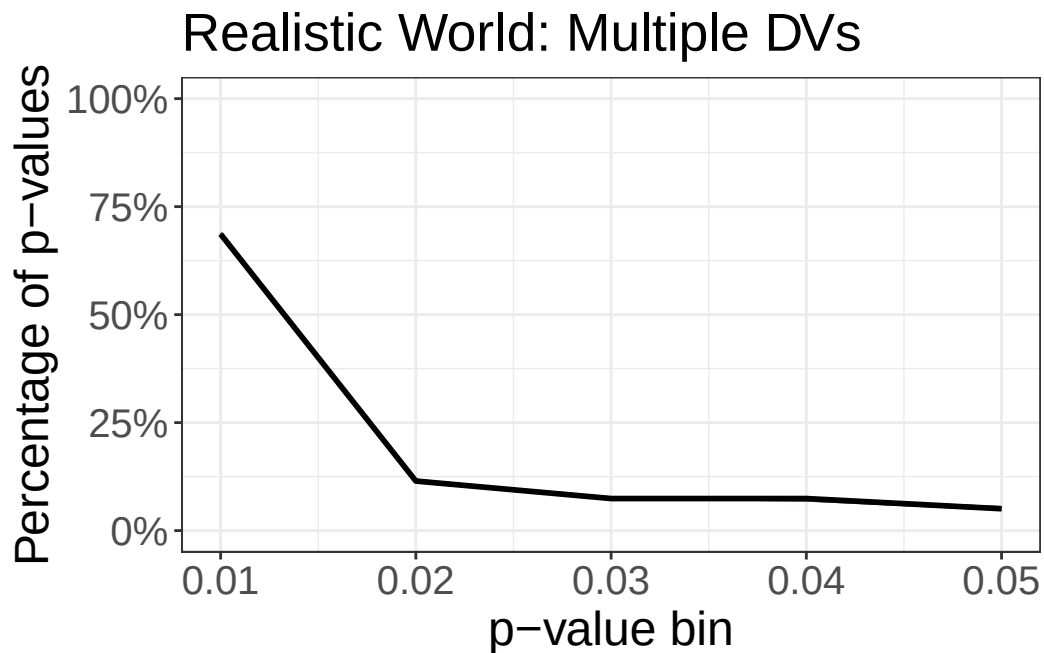
Visualisations of best simulations in each world

You can visually inspect the fit of the top 5 (where available) simulations in each world (in blue) to the target p-curve (in black). Note that the Flat World only involved one simulation.

condition	nvar	r	d	prop_Hacker	prop_H1	het	strategy	nmin	nmax	stepsize	fi
optStop_realistic	NA	NA	0.7	0.25	0.2	0.0	NA	444	1063	100	
multDV_realistic	5	0.3	0.6	0.10	0.3	0.2	1	NA	NA	NA	
multDV_realistic	5	0.3	0.6	0.10	0.3	0.2	2	NA	NA	NA	
multDV_realistic	5	0.9	0.8	0.50	0.2	0.2	1	NA	NA	NA	
multDV_realistic	5	0.9	0.8	0.50	0.2	0.2	2	NA	NA	NA	
optStop_realistic	NA	NA	0.7	0.25	0.2	0.2	NA	444	1063	50	
optStop_realistic	NA	NA	0.8	0.10	0.2	0.2	NA	22	1063	100	
multDV_realistic	5	0.3	0.6	0.10	0.3	0.2	3	NA	NA	NA	
multDV_perfect	1	0.0	0.8	0.00	0.2	0.2	3	NA	NA	NA	
optStop_realistic	NA	NA	0.6	0.25	0.2	0.2	NA	444	1063	100	

Realistic: Multiple DVs

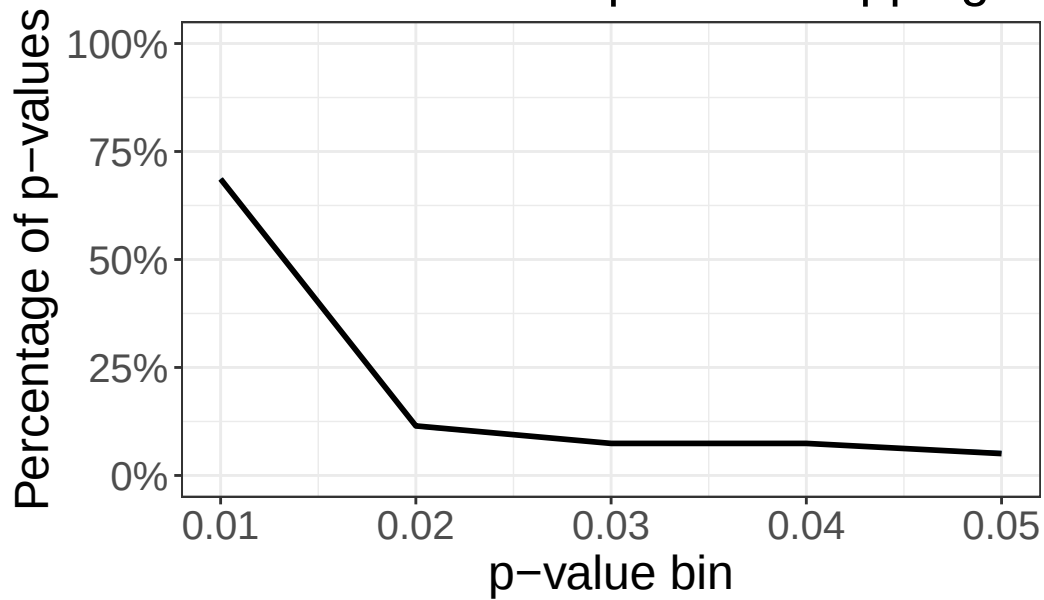
```
plot_pcurves(simres |> filter(condition == "multDV_realistic"), poriginal = simplify2array(p
```



Realistic: Optional Stopping

```
plot_pcurves(simres |> filter(condition == "optStop_realistic"), poriginal = simplify2array(p
```

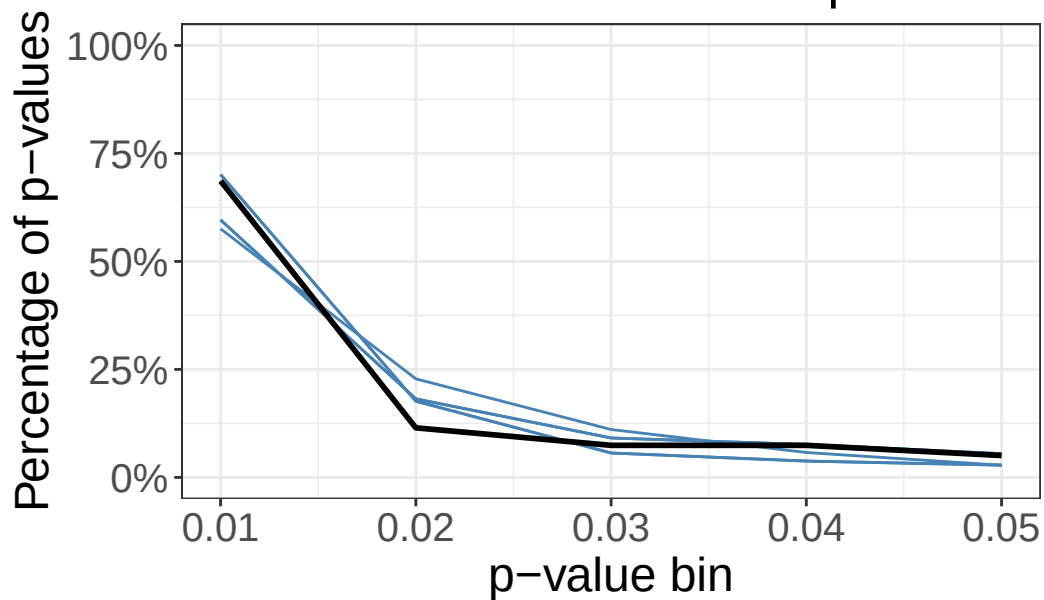
Realistic World: Optional Stopping



Worst: Multiple DVs

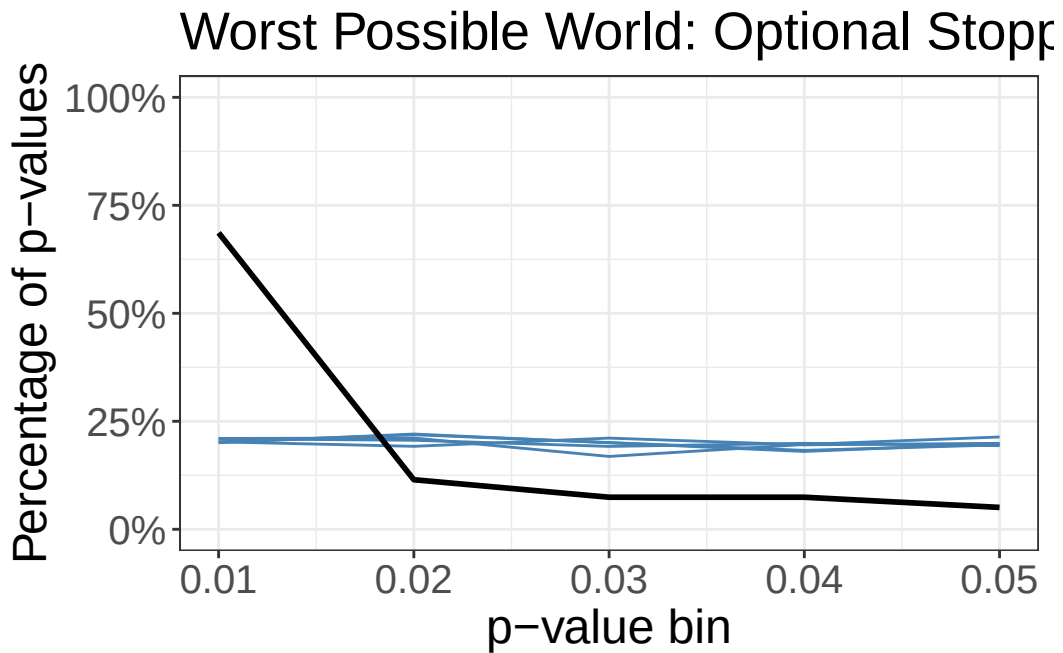
```
plot_pcurves(simres |> filter(condition == "multDV_worst"), poriginal = simplify2array(pcurv
```

Worst Possible World: Multiple DVs



Worst: Optional Stopping

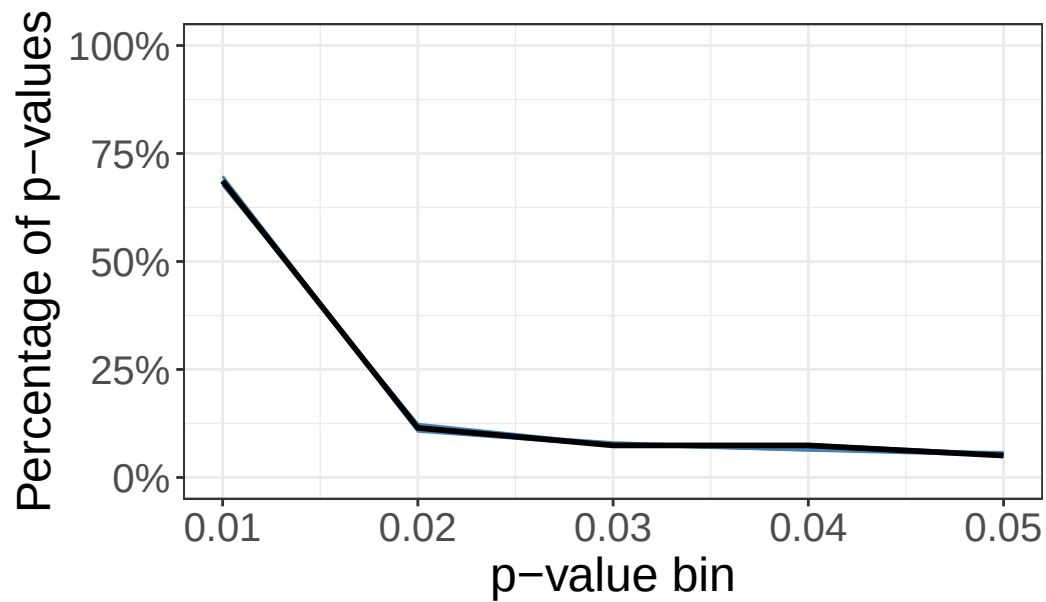
```
plot_pcurves(simres |> filter(condition == "optStop_worst"), poriginal = simplify2array(pcurv
```



Perfect

```
plot_pcurves(simres |> filter(condition == "multDV_perfect"), poriginal = simplify2array(pcurv
```

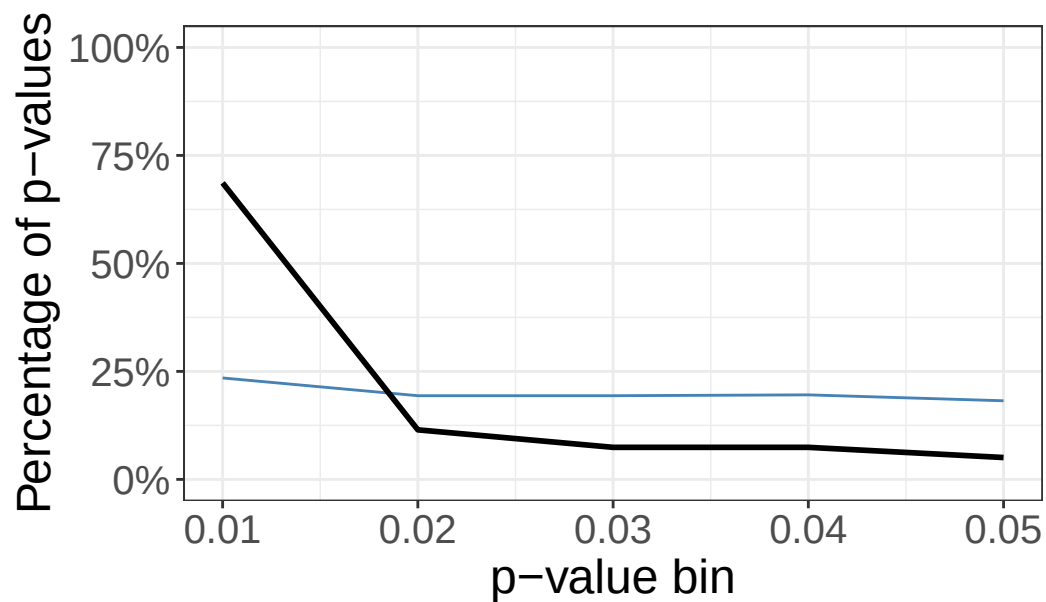
Perfect World



Flat

```
plot_pcurves(simres |> filter(condition == "multDV_H0"), poriginal = simplify2array(pcurves[
```

Flat World



condition	G2	BIC	Chi2	RMSE
Flat World	555.07	555.07	678.65	0.23
Perfect World	0.57	15.85	0.57	0.00
Realistic World: Multiple DVs	0.13	35.78	0.13	0.00
Worst World: Multiple DVs	26.90	47.28	24.91	0.04
Realistic World: Optional stopping	0.17	35.82	0.16	0.00
Worst World: Optional stopping	634.98	655.36	816.17	0.24

Overview of best fitting simulations in each world

These fit indices reflect the fit of the best-fitting simulation in each condition/ world. Lower numbers indicate better fit. Note that BIC is a transformation of G2 that penalises for model complexity.

```
fit_tab <- simres %>%
  group_by(condition) %>%
  summarise(
    G2    = min(fit_wetzels_g2),
    BIC    = min(BIC_wetzels),
    Chi2   = min(fit_wetzels_chi2),
    RMSE   = min(fit_wetzels_rmse)
  )

fit_tab %>%
  mutate(condition = recode(condition, !!!condition_labels)) %>%
  mutate(across(
    -condition,
    ~ cell_spec(sprintf("%.2f", .x), bold = .x == min(.x))
  )) %>%
  kbl(escape = FALSE) %>%
  kable_styling()
```

According to G2, the Realistic World: Multiple DVs contained the best-fitting simulation to the target p-curve. Once model complexity is accounted for, the Perfect World contained the simulation with the best fit to the target p-curve. Results for chi-squared and RMSE are also provided as robustness checks.

What were the top 10 best fitting simulations (according to G^2)?

These rows reflect the 10 simulations that had the best fit to the target p-curve according to G^2 . (Scroll to see the other fit indices for these simulations.)

condition	nvar	r	d	prop_Hacker	prop_H1	het	strategy	nmin	nmax	stepsize	fi
multDV_realistic	2	0.0	0.7	0.50	0.5	0.0	3	NA	NA	NA	
multDV_realistic	2	0.9	0.6	0.25	0.5	0.2	1	NA	NA	NA	
multDV_realistic	2	0.9	0.6	0.25	0.5	0.2	2	NA	NA	NA	
optStop_realistic	NA	NA	0.7	0.10	0.5	0.2	NA	8	444	50	
multDV_realistic	10	0.0	0.6	0.75	0.5	0.2	1	NA	NA	NA	
multDV_realistic	10	0.0	0.6	0.75	0.5	0.2	2	NA	NA	NA	
multDV_realistic	5	0.3	0.6	0.75	0.5	0.0	1	NA	NA	NA	
multDV_realistic	5	0.3	0.6	0.75	0.5	0.0	2	NA	NA	NA	
optStop_realistic	NA	NA	0.5	0.50	0.7	0.4	NA	50	444	100	
multDV_realistic	2	0.9	0.5	0.10	0.7	0.2	1	NA	NA	NA	

```
simres |> arrange(fit_wetzels_g2) |> slice(1:10) |>
  select(condition, nvar, r, d, prop_Hacker, prop_H1, het, strategy, nmin, nmax, stepsize, c
  kbl(digits = 2) |>
  kable_styling() |>
  scroll_box(width = "100%", height = "400px")
```

3. Target p-curve: Simonsohn (2014):

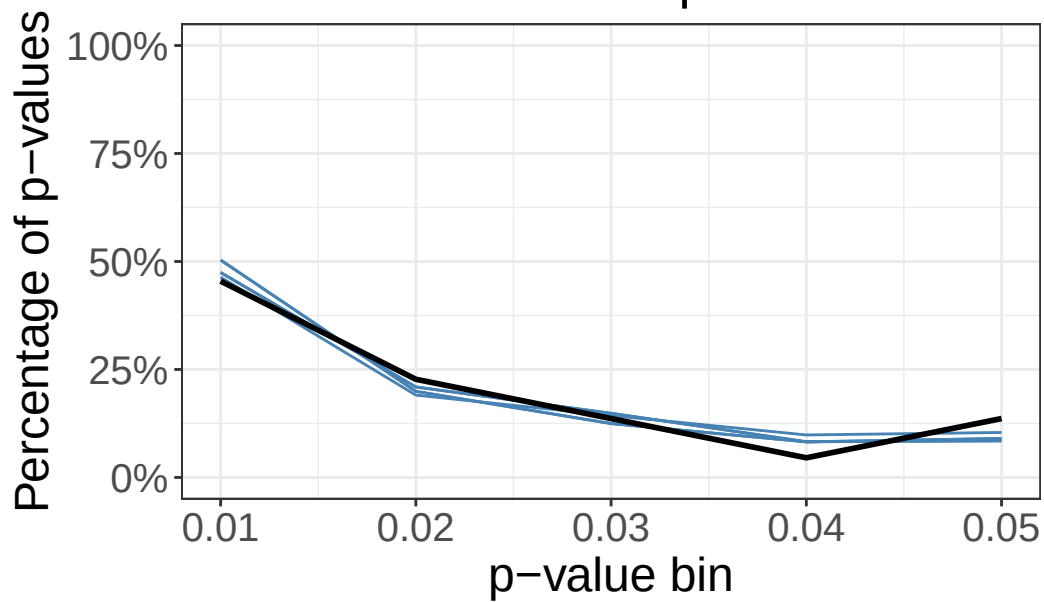
Visualisations of best simulations in each world

You can visually inspect the fit of the top 5 (where available) simulations in each world (in blue) to the target p-curve (in black). Note that the Flat World only involved one simulation.

Realistic: Multiple DVs

```
plot_pcurves(simres |> filter(condition == "multDV_realistic"), poriginal = simplify2array(p
```

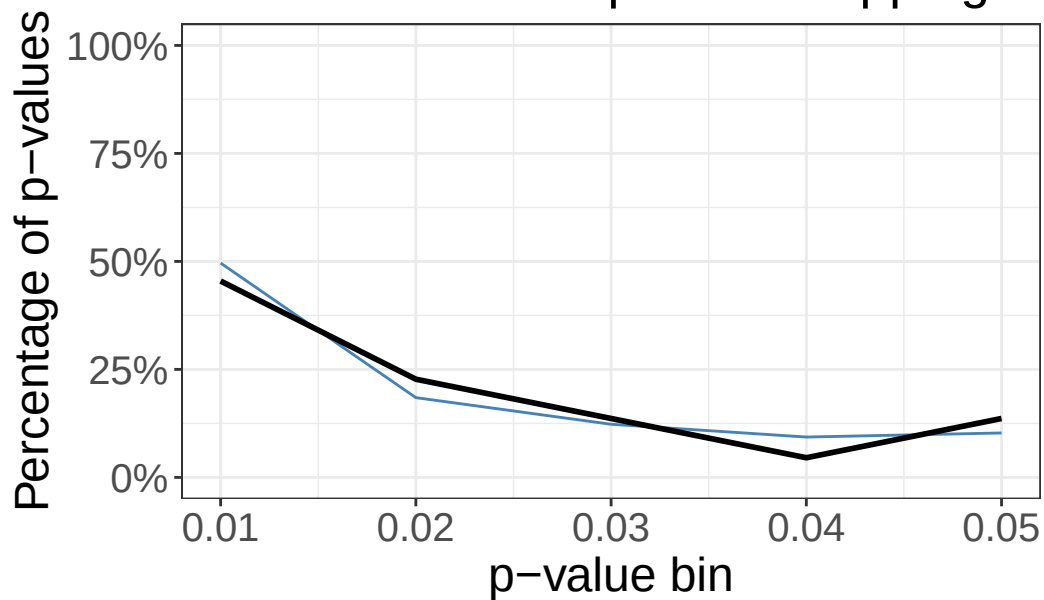
Realistic World: Multiple DVs



Realistic: Optional Stopping

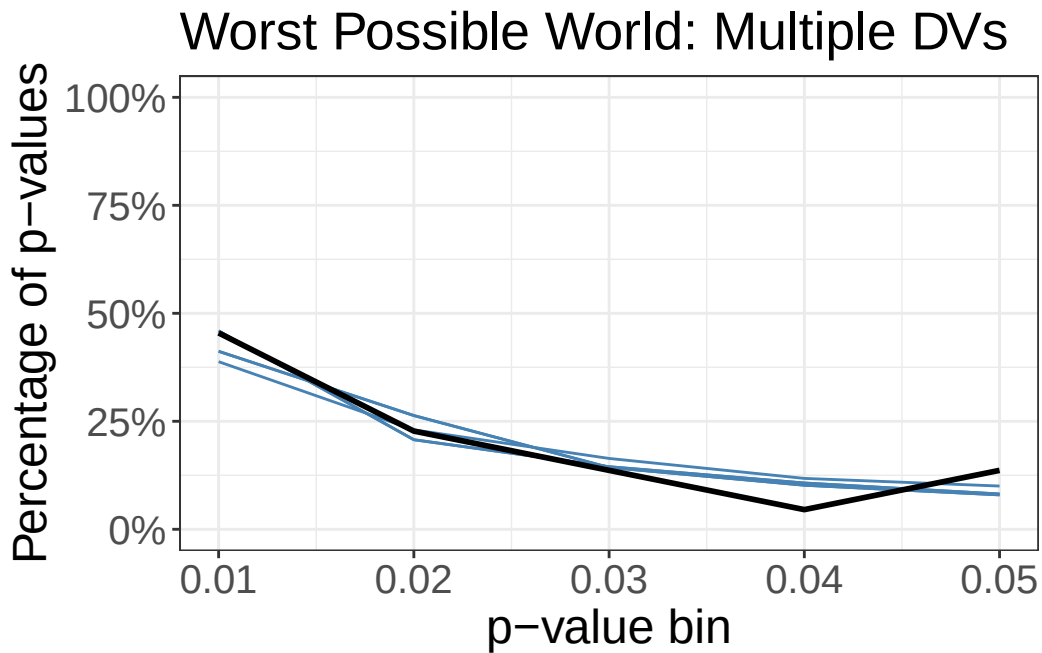
```
plot_pcurves(simres |> filter(condition == "optStop_realistic"), poriginal = simplify2array()
```

Realistic World: Optional Stopping



Worst: Multiple DVs

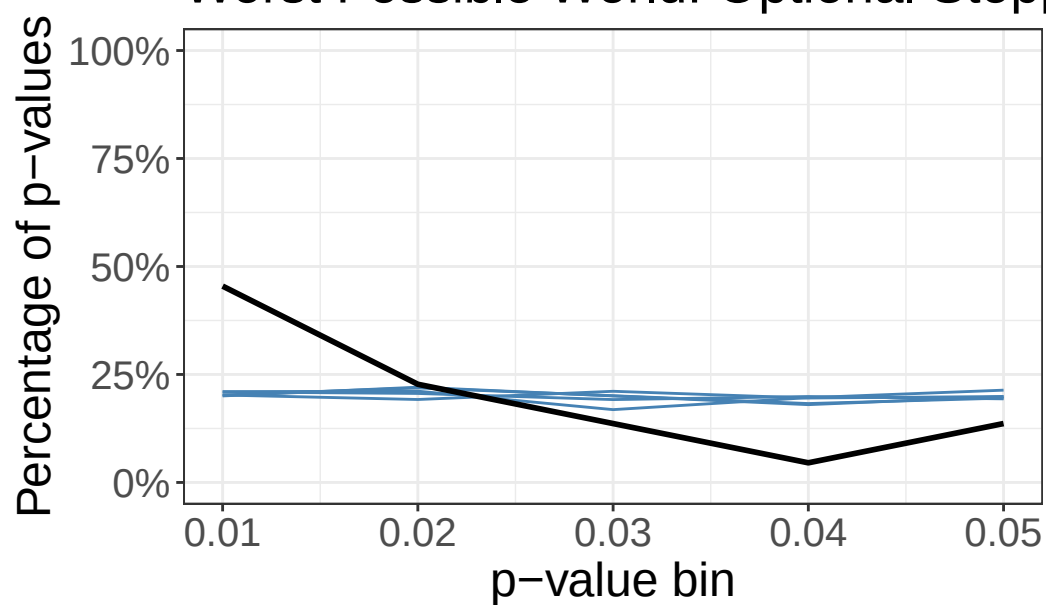
```
plot_pcurves(simres |> filter(condition == "multDV_worst"), poriginal = simplify2array(pcurv
```



Worst: Optional Stopping

```
plot_pcurves(simres |> filter(condition == "optStop_worst"), poriginal = simplify2array(pcurv
```

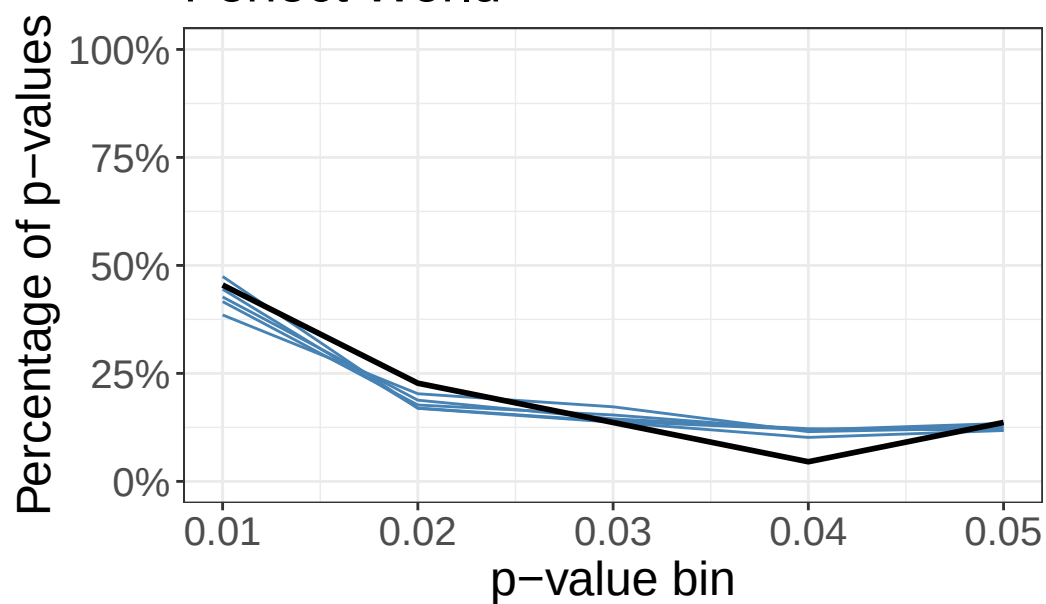
Worst Possible World: Optional Stopping



Perfect

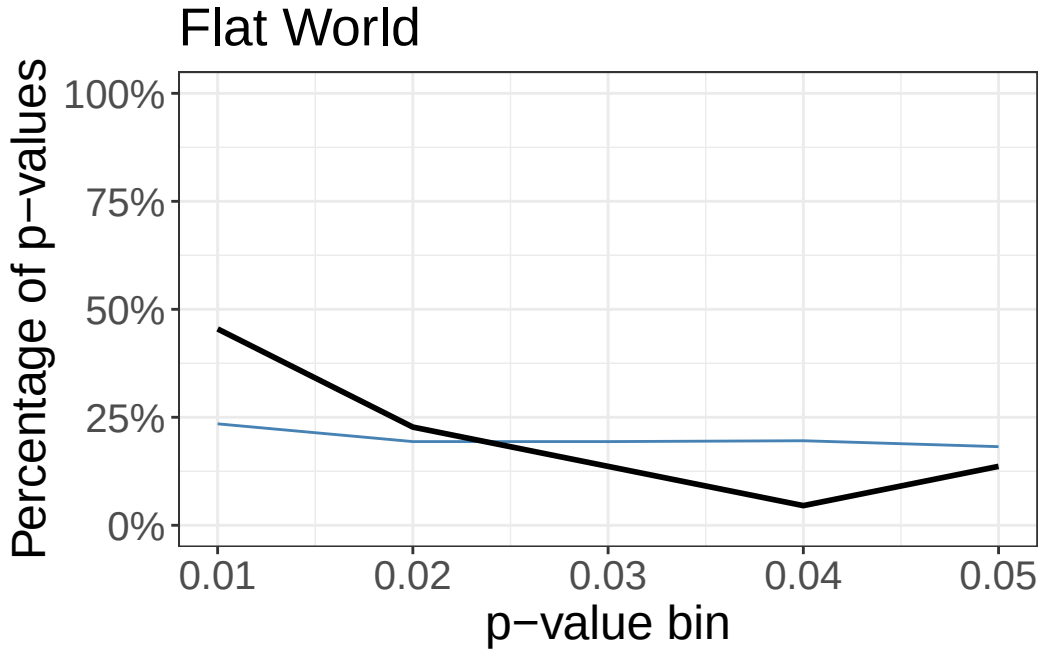
```
plot_pcurves(simres |> filter(condition == "multDV_perfect"), poriginal = simplify2array(pcu
```

Perfect World



Flat

```
plot_pcurves(simres |> filter(condition == "multDV_H0"), poriginal = simplify2array(pcurves[
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Overview of best fitting simulations in each world

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```
fit_tab <- simres %>%  
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    BIC    = min(BIC_simonsohn),  
    Chi2   = min(fit_simonsohn_chi2),  
    RMSE   = min(fit_simonsohn_rmse)  
  )  
  
fit_tab %>%  
  mutate(condition = recode(condition, !!!condition_labels)) %>%  
  mutate(across(
```

condition	G2	BIC	Chi2	RMSE
Flat World	8.05	8.05	7.81	0.12
Perfect World	1.34	16.62	1.21	0.04
Realistic World: Multiple DVs	1.10	36.75	0.97	0.03
Worst World: Multiple DVs	1.72	22.09	1.60	0.04
Realistic World: Optional stopping	1.20	36.85	1.05	0.04
Worst World: Optional stopping	9.26	29.64	9.58	0.13

```

-condition,
~ cell_spec(sprintf("%.2f", .x), bold = .x == min(.x))
)) %>%
kbl(escape = FALSE) %>%
kable_styling()

```

According to G2, the Realistic World: Multiple DVs contained the best-fitting simulation to the target p-curve. Once model complexity is accounted for, the Flat World contained the simulation with the best fit to the target p-curve. Results for chi-squared and RMSE are also provided as robustness checks.

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```

simres |> arrange(fit_simonsohn_g2) |> slice(1:10) |>
  select(condition, nvar, r, d, prop_Hacker, prop_H1, het, strategy, nmin, nmax, stepsize, c
  kbl(digits = 2) |>
  kable_styling() |>
  scroll_box(width = "100%", height = "400px")

```

condition	nvar	r	d	prop_Hacker	prop_H1	het	strategy	nmin	nmax	stepsize	fi
multDV_realistic	70	0.3	0.1	0.25	0.7	0.0	1	NA	NA	NA	
multDV_realistic	70	0.3	0.1	0.25	0.7	0.0	2	NA	NA	NA	
multDV_realistic	70	0.0	0.2	0.10	0.3	0.0	1	NA	NA	NA	
multDV_realistic	70	0.0	0.2	0.10	0.3	0.0	2	NA	NA	NA	
multDV_realistic	150	0.6	0.1	0.25	0.5	0.2	1	NA	NA	NA	
multDV_realistic	150	0.6	0.1	0.25	0.5	0.2	2	NA	NA	NA	
multDV_realistic	2	0.9	0.5	0.25	0.1	0.2	1	NA	NA	NA	
multDV_realistic	2	0.9	0.5	0.25	0.1	0.2	2	NA	NA	NA	
multDV_realistic	70	0.3	0.2	0.10	0.3	0.2	1	NA	NA	NA	
multDV_realistic	70	0.3	0.2	0.10	0.3	0.2	2	NA	NA	NA	